

Compact Substation

English Product Datasheet | Preliminary Project Reference



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Product Overview

Wenze compact substations integrate medium-voltage switchgear, a distribution transformer, low-voltage distribution, internal connections and a protective enclosure. The single-line diagram, transformer technology, protection scheme, ventilation and enclosure system are engineered for the project.

Typical Selection Range

Transformer capacity	315-2500 kVA
High-voltage side	6 / 10 / 11 / 20 / 22 / 33 / 35 kV
Low-voltage side	0.4 / 0.415 kV or project-specific
Frequency	50 Hz / 60 Hz
Transformer type	Oil-immersed or dry-type
HV switchgear	RMU, LBS-fuse or circuit-breaker configuration
LV distribution	Project-specific incomer, busbar and outgoing feeders
Enclosure	Coated steel, stainless steel or composite
Applicable standards	IEC 62271-202 / IEC 60076 / IEC 61439, as applicable
Standard altitude	Up to 1000 m; higher altitude by design review

Important: This document supports preliminary product selection. It does not replace the project-specific approved datasheet, drawings, technical agreement or routine test report.



Project Selection Reference

Typical configurations are shown for inquiry preparation. The destination utility specification, applicable local requirements and site conditions govern final selection.

Parameter	Typical Project Selection	Confirmation Basis
MV interface	Radial or ring-network arrangement	Rated voltage, current and short-circuit duty follow the utility interface.
Transformer section	Oil-immersed ONAN or dry-type AN/AF	Capacity, losses, impedance and vector group are approved separately.
LV switchboard	Incomer, busbar, metering and outgoing feeders	Rated current and short-circuit rating follow the approved single-line diagram.
Enclosure protection	Project-selected IP level and corrosion system	Ventilation, transformer type and site exposure are reviewed together; coastal provisions where specified.
Internal arc	Available where specified	Classification, accessibility type, current and duration require agreement.
Factory scope	Assembly, internal wiring, earthing and agreed routine tests	Site cable termination, civil foundation and commissioning scope are contractual.

Key Product Features

- Integrated MV, transformer and LV compartments
- Compact footprint and reduced on-site assembly
- Project-specific metering, protection and automation options
- Tropical or coastal enclosure provisions where specified and confirmed for the site

Guaranteed Data Confirmed After Design Review

- Approved single-line diagram and equipment schedule
- MV and LV ratings, short-circuit levels and protection scheme
- Transformer guaranteed data and accessories
- Enclosure IP level, ventilation, dimensions and total weight
- General arrangement, foundation plan and cable-entry arrangement

Typical Applications

- Industrial parks and production facilities
- Commercial and residential developments
- Renewable-energy distribution projects
- Mining and construction sites
- Transport and municipal infrastructure

Information Required for Quotation

- Transformer or substation type
- Rated capacity and quantity
- Primary and secondary voltage
- Frequency, phase and vector group
- Destination country and utility specification
- Indoor/outdoor service, ambient temperature and altitude
- Required accessories, protection and metering
- Delivery schedule and applicable standard

Important: Final guaranteed performance, dimensions, weight, accessories and document scope are stated only in the approved project documents issued for the applicable order.